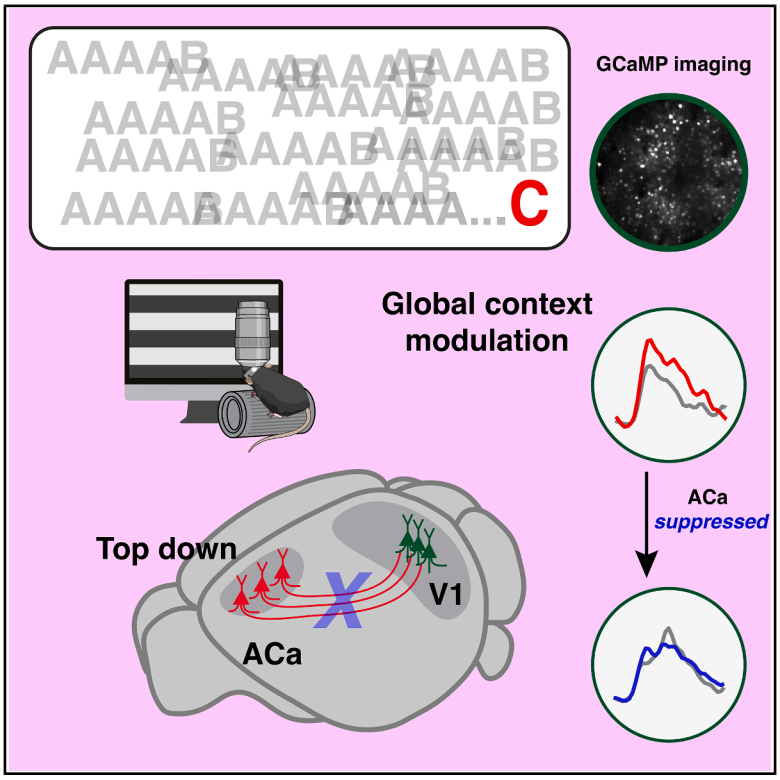
Cell Reports

Anterior cingulate feedback facilitates rapid global context modulation in primary visual cortex

Graphical abstract



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In brief

Peterka et al. show that sensory processing as early as primary visual cortex is modulated not just by recent stimuli but by learned patterns and global probabilities. They go on to demonstrate that this modulation depends on feedback inputs from a higher brain region, consistent with predictive coding hypotheses.

Highlights

- Global context rapidly modulates neural responses in V1
- Global deviations, but not predictable local deviants, elicit enhanced responses
- Downstream brain region ACa is necessary for global context modulation in V1
- Standard oddball paradigms involve higher-order contextual modulation

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Report

Anterior cingulate feedback facilitates rapid global context modulation in primary visual cortex

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SUMMARY

Context modulates sensory processing. Recent proposals claim early areas like primary visual cortex (V1) are only influenced by "local" context (e.g., recent stimulus history), while "global" context (e.g., learned patterns and probabilities) modulates only higher areas. We tested this using a global/local oddball paradigm: mice viewed repeating five-item sequences (AAAA-B), where B is deviant from local context but typical in global context. Using two-photon imaging and local field potential (LFP) recording in mouse V1, we found that global predictability normalized responses to B stimuli. This effect emerged after fewer than 10 sequence repetitions, demonstrating rapid learning. When a globally deviant "C" stimulus replaced B or when B occurred at unpredictable intervals (i.e., standard oddball), V1 showed strong response enhancement. This global context modulation in V1 required top-down feedback from anterior cingulate area. However, when B was replaced by an A (AAAA-A), responses were mixed, suggesting that stimulus-specific adaptation can constrain prediction-error signals in V1.

INTRODUCTION

Sensory processing regions of the cerebral cortex integrate incoming input from the periphery within a broader context, including concurrent and recent stimuli, cross-modal inputs, learned spatiotemporal regularities, and behavioral state. Evidence for such modulation is present as early as primary visual cortex (V1)[1,2] or primary auditory cortex (A1),[3,4] but much of this evidence comes from experimental studies using a standard "oddball" sequence, involving a repeated, redundant stimulus "A" with a rare deviant stimulus "B" interspersed at random times (e.g., AAAAABAAB).

There is ongoing debate as to whether modulation of V1 responses can reflect higher level "expectations," stemming from observed probabilities encoded over longer time scales, or simply the consequences of local adaptation to immediately preceding stimuli—i.e., purely local context.[5] The latter would represent a challenge to some predictive processing theories[6,7]—at least insofar as they hold for V1. A recent report concluded that enhanced V1 responses observed during oddball paradigms reflect responses to deviation (i.e., "deviance detection") only from local context.[8] Specifically, Westerberg et al. used a "global/local" oddball paradigm and found that repeatedly presenting predictable four stimulus sequences such as "AAAB" to a mouse left large V1 responses to "B" intact (local deviant, global redundant), despite the fact that the "B" stimulus was predictable in the global (i.e., longer term) context. Likewise, no such large responses were present in V1 when stimulus "A" replaced the predictable "B" at the end of the sequence (i.e., "AAAA" local redundant, global deviant), although higher cortical areas appeared to detect this deviation. We conclude that (1) this conflicts with the notion that global context can modulate processing in V1 and (2) refutes standard predictive coding theories, which posit top-down suppression of sensory cortical responses to contextually predicted stimuli. Instead, we propose that oddball responses in V1 arise from stimulus adaptation in feedforward circuitry. Similarly, in an auditory global/local oddball study, no A1 responses to local redundant, global deviant "AAAAA" stimuli were present across the neural population, though small increases occur in parvalbumin-positive interneurons.[9] Here, we suggest that global context may be encoded at higher levels of the cortical hierarchy, with weak top-down signals to sensory cortices.

These conclusions are at odds, however, with two lines of work. One, using local field potential (LFP) recordings in V1, has evinced clear deviance detection when complex sequences designed to

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circumvent local adaptation (e.g., ABCD) are violated (e.g., ABDC),[10] yet whether these LFP signals reflect V1 neuronal outputs or distal feedback signals from other brain regions cannot be discerned from LFP alone. Second, other reports studying both V1[1,2] and A1[4] neurons directly show that, even during standard oddball paradigms (i.e., stimuli B occurring at random intervals, e.g., AAAAABAAB), feedback modulation from higher brain regions (e.g., medial prefrontal areas in the mouse) is needed for the full range of context effects—specifically, deviance detection. Importantly, in these standard oddball paradigms,[1,2,4] stimuli B are, indeed, locally deviant just as they are in the AAAB sequences from traditional global/local paradigms, but the timing of stimulus B is not predictable. Thus, in a standard oddball paradigm, B stimuli are both local and global oddballs, complicating comparison between these two types of studies.

One idiosyncrasy of traditional global/local paradigms is that the global deviant in AAA-A is also the locally redundant stimulus.[8,9] That is, replacing the predictable "B" in the AAAB sequence with an "A" creates a condition where neural adaptation (a suppressive effect) and deviance detection (an amplifying effect) are co-occurring yet inverse effects. This complicates interpretations, as our past work has shown that stimulus-specific adaptation (response reductions) to the repeated "A" in a standard oddball sequence is rapid and very strong in V1. Also, such stimulus-specific adaptation (1) is mechanistically distinct from deviance detection, as it depends neither on local SST+ or VIP+ interneurons nor top-down feedback modulation,[1,11] and (2) is present earlier and at lower levels of the visual processing hierarchy than deviance detection (i.e., in layer 4 responses and extracellular currents[12]). Likewise, in the auditory system, stimulus-specific adaptation is present as early as the midbrain and thalamus,[13] several synapses hierarchically below the sensory cortex. Thus, it is possible that the failure to identify global deviance detection using traditional global/local paradigms—and thus, also prediction-error-like responses—in V1 to the "A" stimulus may stem from the particular stimulus design: that this global deviant was presented to a system that had already strongly adapted to the relevant features (i.e., to the "A"), perhaps in synapses prior to the V1 layer 2/3 (i.e., layer 4 or thalamus).

Here, we sought to replicate and then extend the results reported in Westerberg et al. and the theory presented in Gabhart et al.[5] using a similar "global/local" oddball paradigm, through stimuli designed with two additional changes: (1) we presented a condition where the final stimulus in the sequence is both a global and a local deviant ("C") and (2) we included a random orientation control run, where bursts of 5 stimuli are presented of random orientations (Figure 1B). This second change allowed for a comparison of responses to stimuli "A" (90° or 0°), "B" (0° or 90°), and "C" (45°) when they appeared at the end of an unpredictable stimulus chain (e.g., CDAEB; 8 possible orientations, spaced 22.5°) vs. when they came at the end of a predictable stimulus chain (AAAA-x).

RESULTS

Global context modulates V1 responses

We carried out two-photon calcium imaging with GCaMP6s in putative excitatory pyramidal neurons in layer 2/3 of mouse V1 (Figure 1A). We chose to focus on this layer based on our past work showing enriched deviance detection in layer 2/3 relative to deeper layers,[2,12,14] which also conforms to canonical predictive processing models, which suggest that prediction errors are encoded in these layers.[6] Awake mice passively viewed short sets of five consecutive full-field 500 ms oriented moving gratings (100% contrast, 0.08 cpd, 2 cycles per second; 500–600-ms interstimulus interval) separated by a 6-s inter-sequence interval. Mice first viewed 35 repetitions of AAAA-B (habituation run, see Figures 1 and S1), followed by a test run in which B was replaced by C or A every fifth sequence. After this test run, a random sequence control run was presented, lasting 70 sequences of randomly chosen orientations (0–157.5° in 22.5° steps). This process was then repeated (habituation, test, and control), swapping A and B so that the new common sequence was BBBB-A.

We equated trials across conditions and, for the control condition, focused only on A, B, or C orientations appearing in the 4th or 5th position during the control sequence. We first focused analyses on neurons showing statistically significant on-responses (0–500 ms) to a given stimulus orientation (A, B, or C) in the deviant or the control condition (26% to stimulus A; 25% to stimulus B; 33% to stimulus C; see STAR Methods). This was based on Z tests (on-responses vs. pre-stimulus baseline; $p < 0.05$ to both conditions or $p < 0.01$ to either condition) and was done to provide the most accurate estimate of effect sizes by including only neurons with robust responses to a specific orientation (see also STAR Methods and past studies[1,15]). See Figures 2 and S5 for responses across all neurons.

Responses to stimulus "A" were strongly attenuated when it was in positions 1 through 4 compared to the random orientation control (Figure 1C), indicating stimulus-specific adaptation as previously reported.[1,11] Interestingly, when an A deviant replaced the predictable B deviant during the test run, responses to this globally deviant/locally redundant "A" were suppressed, evincing no global deviance detection (Figure 1D; left; $n = 94$ stimulus responsive neurons from 10 mice [6 females], linear mixed-effects model $F(1,177) = -16.9$, $p = 5.820e^{-5}$).

However, we also did not find evidence of true deviance detection to the predictable B deviant in the sequence (Figure 1D, middle; $n = 88$ responsive neurons, $F(1,88) = 0.0004$, $p = 0.988$). While responses to B were not adapted like they were to "A", they were not stronger than responses to B when B appeared in random stimulus sequences (during the control run).

On the other hand, when a different global deviant ("C") was presented; which was not adapted by the preceding A stimuli (i.e., 45°), or by the global context (the predictable B deviant), a strong and significant deviance detection signal was present, relative to the control run (Figure 1D, right; $n = 121$ responsive neurons; $F(1,121) = 9.750$, $p = 0.002$).

A similar pattern of effects was present in LFP recordings in V1 to B and C deviants in a separate set of mice ($n = 9$, 5 females; Figures 1E and 1F; factorial ANOVA on theta (4–12 Hz) and gamma (20–40 Hz) power,[16] 75–600 ms post-stimulus onset, context × deviant-type interaction, $F(3,32) = 4.35$, $p = 0.011$), again with significant deviance detection signal to the "C" global deviant ($t(8) = 2.35$, $p = 0.023$; one-tailed, significant after false discovery rate [FDR] correction) but not the "B" local-only

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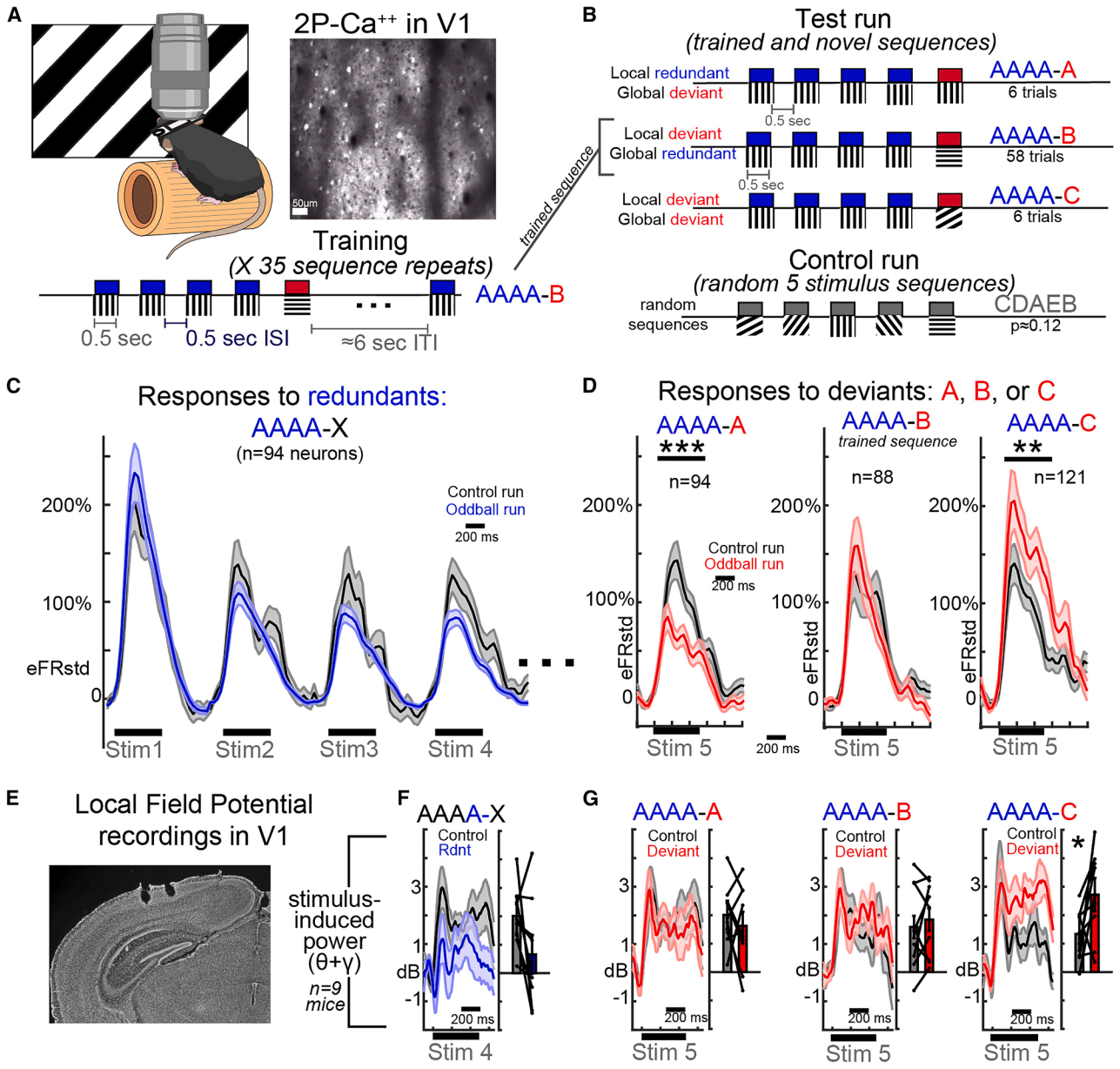


Figure 1. Deviation from global context is detected in V1 but obscured by stimulus-specific adaptation

(A) Head-fixed mice (n = 10; 6 females) viewed full-field moving, oriented grating stimuli while V1 neurons (n = 360 total) were recorded with two-photon calcium imaging (GCaMP6s).

(B) Mice viewed 5 stimulus sequences (4.5-s duration) of AAAAB, presented 35 times for habituation.Scale bar is 0.05 mm. Then, after a short (<3 min) break, mice viewed a test run during which the final "B" in the sequence was occasionally replaced with an A or a C. This was followed by a control run, during which randomly orientated gratings were presented in similar 5 stimulus sequences.

(C) Stimulus-evoked activity of neurons responsive to the A stimulus during the control run (gray; when in the 1st, 2nd, 3rd, or 4th position) or the test run (blue).

(D) Stimulus-evoked activity of neurons responsive to A, B, or C stimulus when that stimulus was deviant (red) or in the 4th or 5th position in the control run (gray). Magnitudes reflect standardized Caiman-estimated firing rate (eFRstd), scaled to average of control response (0–500 ms) for comparison across plots.

(E) Bipolar electrodes inserted into V1 (200–400 μm below surface) recorded LFP during the same paradigm.

(F and G) Morelet wavelet-transformed stimulus-induced power, averaged across theta (4–12 Hz) and gamma (20–40 Hz) frequencies for the same conditions as in (C) and (D). One observation per mouse. Linear mixed-effects model F statistics for neurons or paired t tests for LFP. *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$. Error bars reflect standard error of the mean across neurons (C and D) or across mice (F and G).

deviant ($t(8) = 0.65$, $p = 0.267$, one-tailed). Deviance detection was also absent to the A stimulus was in the 5th location in the LFP (globally deviant; $t(8) = 0.77$, $p = 0.463$; two-tailed). Interestingly, stimulus-specific adaptation was also absent from this A, in contrast to the single neuron effects (Figure 1C). Effects did not differ between sexes.

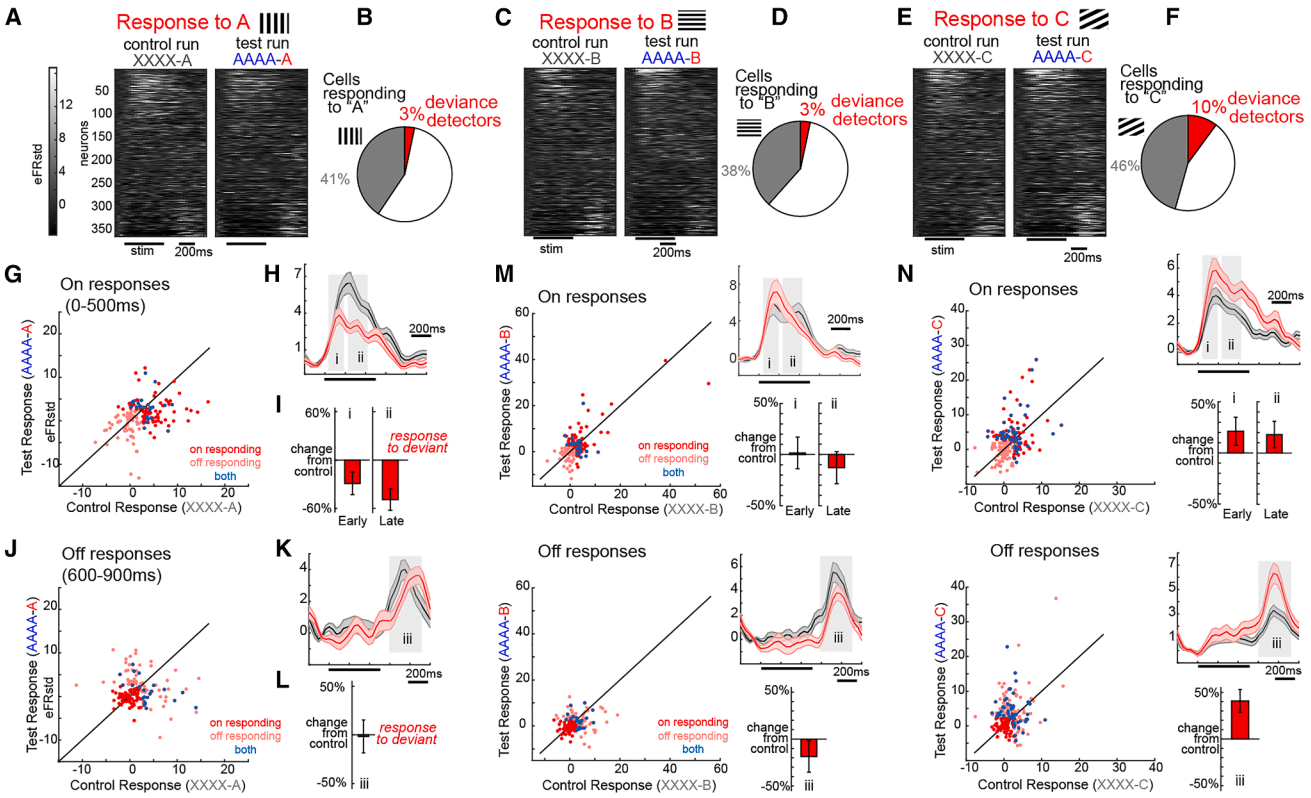


Figure 2. Global context impacts early and late latency on-responses as well as off-responses in V1

(A) Average responses in all recorded neurons to A-oriented stimuli.

(B) Proportion of neurons responsive to the orientation (gray) and statistically significant deviance detection (red; one-tailed *t* test, $p < 0.025$).

(C and D) same as (A) and (B), but for B orientation.

(E and F) same for C orientation. Neurons are aligned across all raster plots for cross-condition comparisons.

(G) Scatterplots of average on-responses (0–500 ms) to stimulus A when it was in the control (*x* axis) vs. deviant context (*y* axis). All neurons are displayed, with significantly responsive neurons colored according to the legend.

(H and I) Same as Figure 1D, but indicating time windows for (I), bar plots of on-responsive neurons scaled to control response, for early 0–175 ms (i) and late 210–350 ms (ii).

(J–L) same as (G)–(I) but for off-responses.

(M and N) same as (G)–(L) but for stimulus B and C. Error bars reflect standard error of the mean across neurons.

Thus, deviance detection was present only to stimuli that were neither locally redundant (the "A") nor globally predictable (the "B"). Interestingly, during the initial habituation run (35 presentations of AAAAB or BBBBA), deviance detection responses were, in fact, present to the B in the first ≈7 trials, but this attenuated rapidly and was completely absent by the end of the habituation (Figure S1). Similar rapid contextual learning in V1 has been previously reported in spatial navigation tasks.[17,18] Conversely, stimulus-specific adaptation to A stimuli was present throughout (Figure S1).

Next we examined the distribution of responses of individual neurons to each of the deviant types (Figures 2A–2F). Past reports show that deviance detection, like sensorimotor prediction errors, is not expressed ubiquitously throughout layer 2/3 neurons but in a subset of mainly pyramidal neurons.[1,2,14,19] In line with this finding, we found that about 10.2% of all active pyramidal neurons showed deviance detection to stimulus "C" (or ≈18.3% of all neurons responsive to C; Figure 2F; one-tailed paired-samples *t* test with $p < .025$ between control and deviant across trials), consistent with past estimates for passive visual prediction errors. These estimates were considerably lower to the A (3.3% of scored/active neurons ($n = 362$) showed deviance detection) and B deviant types (also 3.3%; Figures 2B–2D), and were visually scarce upon inspection (Figures 2A–2C and 2E).

We next split neurons into on-responsive (Figures 2G–2I; 0–500 ms) and off-responsive neurons (Figures 2J–2L; 600–900 ms). Stimulus responsive neurons exhibited either only on-responses (33%), only off-responses (30%), or responses to both (37%; Figures 2A–2C and 2E). For on-responses, about 53% of responsive neurons exhibited responses to only one orientation, while only 11.5% exhibited responses to all three. This selectivity was less pronounced for off-responses, for which only 34% of PYR neurons showed selectivity for a single orientation (for a full characterization, see Figure S2). We next examined effect sizes across early (0–175 ms), late (210–350 ms), and post-stimulus periods (600–900 ms) for each of the main effects. This was done in an exploratory manner mainly for understanding the temporal patterns of deviance processing for building future

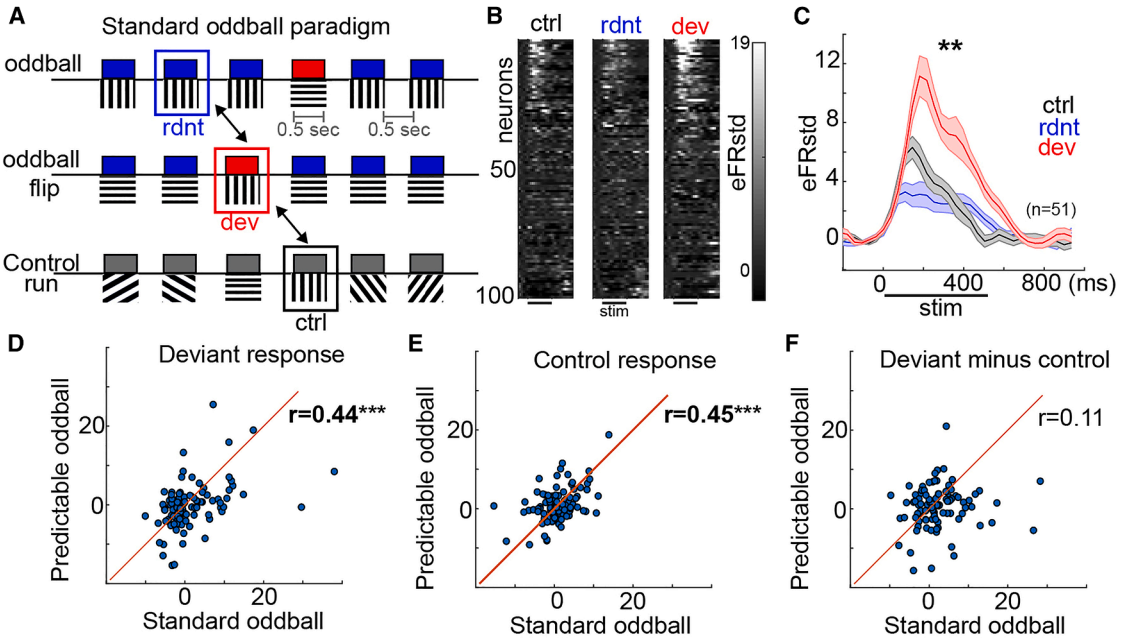


Figure 3. Deviance detection responses during the standard oddball reflect global context processing

(A) Schematic of standard oddball paradigm, using the same features, duration, and interstimulus interval as the global/local, but continuous presentation and no predictability of oddball timing. Comparisons are made to a many-standards control sequence (bottom) where 1 of 8 oriented stimuli is presented at random, such that each orientation (stimulus B) is equally probable as in the oddball run.

(B) Average responses of recorded V1 layer 2/3 neurons from 3 mice to the same orientations when it was neutral, redundant, or deviant.

(C) Stimulus-evoked activity of neurons responsive to one of the two oriented stimuli (n = 51; LME, $^{**}p < 0.01$).

(D) Comparing responses in each neuron to the oddball (deviant) orientations during the standard oddball (x axis) vis the predictable oddball (i.e., the B when AAAAB was the habituated sequence).

(E) Same as (D) but for the control run.

(F) Same as (F) but for the computed deviance detection score of deviant minus control. Pearson correlations; $^{***}p < 0.001$and FDR-adjusted $q < 0.05$. Error bars reflect standard error of the mean across neurons.

hypotheses, rather than drawing strong inferences about stages of processing. As described above, on-responses to the deviant A were attenuated, suggesting dominant stimulus-specific adaptation and the absence of global deviance detection. This effect was present for both early and late stimulus periods (early Cohen's $d = -0.21$; late $d = -0.55$). On the other hand, no such suppression was present in off-responses, nor was there significant deviance detection, to the final A (Figures 2J and 2K; off-d = −0.13). To the B deviant (i.e., the ''predictable'' deviant), deviance detection was absent (Figure 2M) and this pattern held true for early (d = 0.03) and later ($d = -0.09$) response windows, compared to the control condition. On the other hand, off-responses to B were attenuated relative to the control (Figure 2M, bottom; $d = -0.33$). Conversely, deviance detection to the C deviant was present in early (d = 0.24), late (d = 0.27), and off time windows (d = 0.48). The fact that this is present early suggests that deviance detection to the C stimulus is not necessarily fed-back from higher brain regions, but is present in early spiking responses to the deviant C, in line with the notion of ongoing predictive modulation occurring at—or prior to—the onset of stimuli.[1] Still, the overall pattern of off-responses suggests even more dramatic global context effects at these later latencies, as A and C off-response effect sizes (Figure 2Kii) were augmented relative to the peri-onset effects (Figure 2Hi), while B responses were statistically attenuated. This suggests that later-occurring feedback may additionally enhance global context processing beyond the modulation that is present at the initial stimulus evoked response, as suggested by Furutachi et al.[17,18]

Robust deviance detection in a standard oddball sequence suggests global context processing

The most common paradigm for studying context processing and predictive coding in sensory areas is the standard oddball paradigm, involving randomly inserted deviant "B" stimuli in a train of redundant "A" stimuli (Figure 3A). Past work[1,2,11,12] shows that deviant B stimuli induce a strong response in V1 that is distinct from stimulus-specific adaptation, both in magnitude (compared with a random-sequence control) and in mechanism (dependent on top-down feedback and local interneurons). However, in this standard oddball paradigm, deviant B stimuli are both local oddballs (i.e., different from the immediately preceding stimuli) and global oddballs (differing from global probabilities), making it unclear whether deviance detection responses to B stimuli reflect local or global context.[5] In a subset of mice, we directly compared responses to B stimuli (0° or 90°) when they presented in a standard oddball vs. a many-standards control sequence, wherein B stimuli were equally probable as they were during the

oddball run, but not contextually deviant, as multiple orientations were presented without any repetition or structure.

As expected, B responses in the standard oddball were dramatically larger than they were to the control, suggesting deviance detection that was distinct from simple stimulus-specific adaptation (Figures 3B and 3C; $n = 51$ responsive, matched neurons, LME $F(1,100) = 3.59$, $p = 0.001$). When examining the same neurons responding to "B" during the standard oddball vs. the predictable (global/local) oddball, response magnitudes were correlated between the two deviant ($r = 0.44$; $p < .0001$) and the two control runs ($r = 0.45$; $p < 0.0001$) as expected. However, these standard oddball deviance detection responses (deviant minus control) were considerably larger than responses in the same neurons to the deviant-B stimulus when it was *globally* predictable (i.e., during the AAAA-B sequences; all matched neurons $t(98) = 2.14$, $p = 0.033$) and, curiously, were not correlated ($r = 0.11$; $p = 0.276$), suggesting that deviance detection in the standard oddball is distinct from the local context modulation present in the global/local paradigm.

In sum, the standard oddball paradigm, used for decades to study basic sensory-cognitive function in health and neuropsychiatric disease,[20,21] evokes deviance detection responses that reflect at least some degree of global context processing (i.e., beyond a change from the immediately previous stimulus, e.g., global probabilities). Further, the timing of a deviant stimulus in a standard oddball paradigm is not predictable, in contrast to the AAAAB sequences to which mice were habituated here. In the auditory system, past work has shown that randomly timed oddballs elicit larger responses than periodic (temporally predictable) oddballs[22]—a point that is in line with our contrast between the standard oddball and the local AAAAB oddball. Critically, isolating this deviance detection requires the use of appropriate contrasts in the analyses (i.e., deviant vs. control, rather than deviant vs. redundant).

Global context processing in V1 depends on top-down feedback from higher brain areas

Finally, past work has shown that basic deviance detection in early sensory areas during the simple oddball paradigm depends on top-down feedback to V1[1,2] or A1[4]. Here, we tested whether such feedback is also necessary for global context modulation seen during the global/local paradigm. We focused on anterior cingulate area (ACa), which has previously been shown to densely innervate V1 in a manner that supports context processing in a variety of paradigms.[1,23–25] Specifically, ACa densely innervates layer 1 of V1,[2,26] synapsing on multiple interneuron types as well as apical dendrites of excitatory neurons.[24] Functionally, this impact is strongest on VIP+ interneurons,[1] which are critical for sensory deviance detection[27,28] and mediate both direct inhibition and spatiotemporally precise disinhibition of excitatory neurons in V1.[29]

We focused on the pattern of effects observed in Figures 1 and 2, i.e., absent deviance detection to the globally predictable "B" and present deviance detection to the globally deviant "C", and we tested whether this modulation depended on ACa activity. Thus, our main analyses focus on these conditions (responses to A deviants and redundants are presented in Figure S3). We targeted this feedback circuit using a viral strategy (Figure 4A), expressing inhibitory DREADD receptors (hM4Di) in excitatory neurons of ACa, followed by delivery of DCZ (a DREADD ligand) or saline on separate days (Figures 4B and 4C). First, we carried out LFP recordings in both ACa and V1 to confirm both DREADD suppression of ACa and evaluate ACa's role in V1 context processing at a regional level. DREADD activation with DCZ suppressed ACa activity (high-gamma, 80–200 Hz power; LME with order (day 1 vs. 5) included as a fixed-effect variable, $t(14) = -3.24$, $p = 0.006$) but not V1 ($t(14) = -1.27$, $p = 0.223$; Figures 4D and 4E). This suppression of ACa abolished contextual modulation in V1 during the global local oddball paradigm (LME; 3-way interaction of drug by context by deviant-type, $F(1,84) = 4.83$, $p = 0.031$). On the day with saline injection, mice exhibited clear deviance detection to the global C deviant (Figure 4F, $t(11) = -3.26$, $p = 0.004$, FDR-adjusted $q < 0.05$) but not the predictable B deviant ($t(11) = 1.299$, $p = 0.110$). On the day with DCZ, mouse LFPs did not exhibit deviance detection in any condition (Figure 4G; C deviant, $t(11) = -0.392$, $p = 0.350$; B deviant, $t(11) = 0.45$, $p = 0.335$). ACa suppression did not dramatically alter responses in the control condition or to A stimuli, when redundant or deviant (Figure S3). Thus, the impact of ACa suppression was to reduce the relative strength of responses to global (C) vs. local (B) deviants, and this was robust across mice, regardless of the order of DCZ vs. saline injection days (Figure 4H).

Next, we repeated these LFP experiments in a second cohort of mice ($n = 10$, 2F) via two-photon calcium imaging in layer 2/3 neurons, as in Figures 1, 2, and 3. We expressed hM4Di DREADDs in ACa in 10 mice, while expressing GCaMP8m ($n = 5$) or GCaMP6s ($n = 5$) in V1 (Figures 4I–4K). We repeated the same approach as with the LFP and observed again that suppression of ACa abolished contextual modulation in V1 during the global local oddball paradigm (LME; 3-way interaction of drug by context by deviant-type, $F(1,84) = 4.43$, $p = 0.035$; accounting for order (saline day or DCZ day first) and indicator (GCaMP8m vs. 6s) as fixed effects; mouse, field of view, and cell as random effects; see Table S1). As we saw in our other experiments, both LFP and calcium imaging, saline-treated mice exhibited strong deviance detection to global C deviants ($F(1,301) = 7.07$, $p = 0.008$, FDR-corrected $q < 0.05$; Figure 4M) but not local B deviants ($F(1,140) = 1.60$, $p = 0.208$; Figure 4L). In contrast, DCZ treatment abolished deviance detection to the global C deviants ($F(1,316) = 0.006$, $p = 0.936$, Figure 4O), as we saw in the LFP recordings (Figure 4G). However, ACa suppression also led to strong deviance detection to the predictable, local B deviant ($F(1,248) = 10.45$, $p = 0.001$)—an effect that was not apparent in the LFP data (discussed further). Above all, this suggests that top-down modulation of V1 may involve some suppression of local deviance detection. This bidirectional modulation is consistent with past findings during the oddball paradigm,[1,2] suggesting not a simple top-down inhibition or augmentation of V1 during visual processing, but a context-dependent modulation of response gain.

DISCUSSION

These results demonstrate that global context *can* modulate early visual cortical activity, yet provide mixed support

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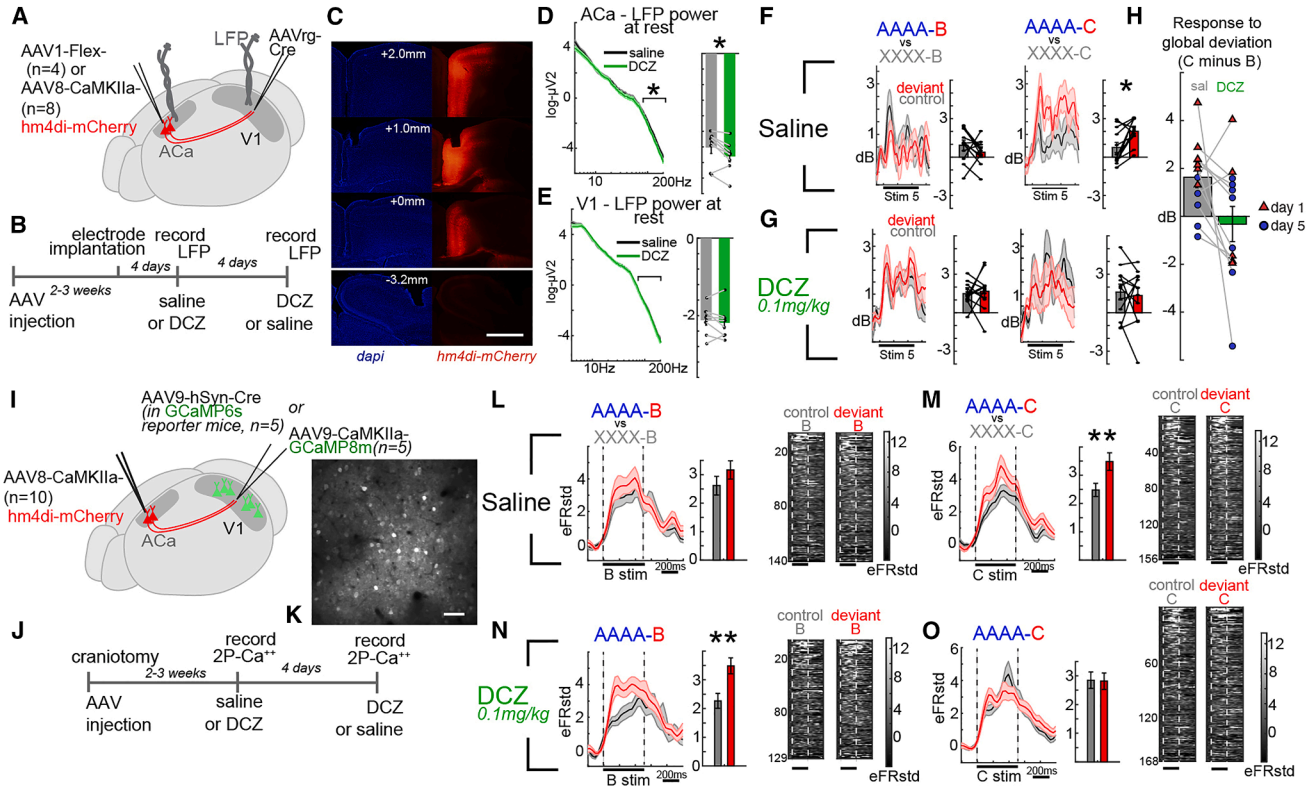


Figure 4. Global context processing depends on top-down feedback to V1

(A) AAVs targeting Aca-to-V1 projections with inhibitory DREADDS (hm4Di) were injected into Aca, and bipolar electrodes were inserted into ACa and V1 in 12 mice.

(B) Two weeks later, LFPs were recorded during the global/local oddball paradigm after mice were given 0.1 mg/kg DCZ (to inactivate Aca-to-V1 projections) or an equivalent volume of saline i.p. (intraperitoneal). Five days later, mice were given the other treatment (saline or DCZ), and LFPs were recorded again.

(C) Electrode location and hm4Di expression were confirmed post hoc. Scale bars, 1mm.

(D and E) DREADDs effectively decreased high-gamma LFP power in ACa (80–200 Hz; $n = 8$ mice, subset with confirmed ACa electrodes) (D) but not V1 (E).

(F) After saline injections, mice exhibited global deviance detection to C but not B, replicating Figure 1F (theta/gamma power, 75–400 ms post-stimulus onset, averaged over trials).

(G and H) When ACa was suppressed with DCZ, contextual modulation during the paradigm was absent (G). This effect was exhibited in (H) reduced global vs. local oddball responses after DCZ (C minus B; statistics were not performed on (H) as they are not independent of that in (F)–(G).

(I) Inhibitory DREADDs (hm4Di) were expressed in Aca-to-V1 projections, as in (A), and GCaMP6s ($n = 5$) or GCaMP8m ($n = 5$) was expressed in V1, followed by 2 weeks of recovery and recordings as in (D)–(G).

(K) Excitatory neurons expressing GCaMP8m in V1 layer 2/3. Scale bars, 0.05 mm.

(L) For the saline condition, to the B deviant line plots (averaged over responsive neurons, $n = 140$; and trials, $n = 4$–6 per condition), bar plots of data from (L), averaged from 0 to 500 ms, and a raster plots of average responses of data from (L), separated by neuron.

(M) Same as (L) but for C deviant.

(N and O) Same as (L) and (M) but for DCZ-treated mice. Linear mixed-effects model F statistics, $^{*}p < 0.05$ and $^{**}p < 0.01$. Error bars reflect standard error of the mean across mice (D–H) or across neurons (L–O).

for predictive processing theories. Prediction-error-like responses were absent in V1 to globally predictable B stimuli, but they were present to both the globally and the locally deviant C, in support of predictive coding theories. The higher-order context—that a B was predictable at the end of each sequence—was learned rapidly, within the first exposure session, and modulated early visual cortical responses at the neuron level. Context-dependent response modulation to C vs. B in these sequences was dependent on higher cortical areas that provide feedback to V1, namely, ACa. This point is also consistent with predictive coding theories.

However, although deviance detection was absent from predictable B stimuli, responses to predictable B were also not suppressed relative to the contextually neutral B. These findings suggest that, in this passive-viewing paradigm, top-down feedback in predictable contexts provides a gain-modulation role. That is, it serves to amplify responses to deviants, rather than to suppress responses to redundants (at least in layer 2/3), consistent with a top-down disinhibition model.[1] The absence of an overt suppression of neural responses to B (relative to the controls) should not be taken as evidence against predictive suppression under all circumstances, however, given strong existing evidence for top-down inhibitory suppression in sensory

cortices under paradigms involving motor-sensory, audiovisual, and auditory-auditory contingencies established over longer time scales.[14,25,30–32] Alternatively, predictive suppression could occur in downstream brain areas (e.g.,[15]), or it could emerge later, after days of exposure, as seen in the aforementioned studies.[14,25,30–32]

An important caveat to this primarily top-down disinhibition model is that during chemogenetic suppression of ACa, deviance-detection responses to the predictable B reemerged in V1 layer 2/3 neurons (Figure 4N). This suggests that other mechanisms, either local to V1 or in other top-down projecting regions, may support more local deviance detection (i.e., augmentation [beyond the neutral control] to Bs preceded by As), and that this augmentation is suppressed by ACa input to V1 (which would also be a form of predictive suppression, see above). Yet, the C deviant was both local and globally deviant, and so it is curious why C deviants no longer elicited deviance detection after ACa suppression (Figures 4G and 4O). One possible resolution stems from the fact that our B stimuli were 90° different from the preceding A, while C stimuli were only 45° different. Figure S2 demonstrates that very few neurons showed responses to both B and A, while nearly 3 times as many showed responses to both C and A—an observation in line with what is known about tuning curves in V1 neurons.[33] Altogether, this would mean that (1) local deviance detection is sensitive to certain stimulus features (degrees of difference from the immediately previous stimuli), and (2) this form is suppressed by ACa input to V1; further, (3) global deviance detection is less sensitive to these stimulus features, and (4) is also supported by ACa input to V1. Future work can help tease out these details, which, though granular, may be crucial for fully evaluating and refining theories of predictive processing in the neocortex.[34]

Our findings are consistent with canonical, precision-weighted formulations of predictive coding[35,36] and closer to a top-down disinhibition model. In such formulations, superficial pyramidal neurons are thought to encode precision-weighted prediction errors,[6,37] where precision reflects the confidence assigned to a given prediction error and is formally analogous to the Kalman gain in Bayesian filtering treatments of predictive coding.[38] Neurophysiologically, this precision weighting is typically cast as a modulation of postsynaptic gain, arising from fast, reciprocal interactions between superficial pyramidal cells and inhibitory interneurons.[39,40] Under hierarchical predictive coding, such precision-weighting can be implemented via top-down neuromodulatory feedback that shifts with context: repeated exposure to a predictable sequence would allow the system to anticipate an upcoming event and adjust its attentional or gain "set" pre-emptively, prior to stimulus onset, rather than only correcting for surprise after the fact. This has implications for interpreting our results: an increase in response amplitude (i.e., to the "C" deviant) could equally reflect a genuine violation of expectation (a prediction error proper) or a context-driven shift in precision-weighting that alters gain irrespective of whether an expectation was violated. Our data cannot fully adjudicate between these two possibilities, and we raise this distinction here because it bears directly on the anticipatory, tonic role we ascribe to ACa feedback above and have observed in the past.[1,2]

Standard deviance detection was absent from the globally deviant A in V1—at least in the firing output of layer 2/3 neurons. As noted earlier, this need not challenge predictive coding broadly, as each cortical area may compute prediction errors only from the predictive and sensory information available to it.[15,41] Our data point to a specific version of this: repeated exposure to "A" drives strong bottom-up adaptation that dominates V1 responses under conditions of high redundancy, curtailing the information available for prediction-error computation by the time signals reach layer 2/3.

Past work has shown that signatures of stimulus-specific adaptation are present both at earlier latencies and further upstream than signatures of deviance detection.[12,13] It is possible that if certain stimulus features are already adapted sufficiently prior to reaching deviance detection computing circuitry in layer 2/3, deviance detection may be dampened or altogether eliminated in V1. That said, another possibility is that mice do not perform true counting.[42] That is, the internal model learned in this paradigm could be that of a B coming after some approximate but unquantified string of A stimuli. Therefore, an A in the 5th spot does not trigger a prediction error because the mouse cortex does not differentiate the 5th A from the 4th A. Future work could test this by inserting a B stimulus into the 3rd or 4th position. A deviance detection response to the B in the 4th or 3rd position, but not the 5th, would clarify the nature of the internal predictions.

Our main analysis chose to focus on responsive neurons with significant responses to a given orientation (0°, 45°, or 90°) across conditions. On average, this was 24.3% of neurons across comparisons (range: 19.4%–33.4%; see Table S1). While this helped us to be confident in estimating effect sizes of each form of deviance detection, the pattern of effects is unchanged if we include all recorded neurons in our analyses (Figure S4; splitting by neurons showing positive vs. negative responses to stimuli; combined across all mice, excluding DREADD-expressing mice on DCZ days).

Interestingly, our experimental results replicate those of a recent study by Westerberg et al., though our overall conclusion differs, with our interpretation informed by our extended experimental design. We find that higher-level context modulation and complex deviance detection is present in mouse V1. The Westerberg study reported augmented responses to the local oddball (i.e., the globally predictable B in the learned AAAB sequence) when compared against the third "B" in a BBBB control condition, and ascribed that to a true deviance response. Ostensibly, our data are also consistent with this pattern (see Figure 1C vs. 1D), but we argue that a more appropriate contrast is a "B" stimulus present under control conditions where orientations are mostly unpredictable—and which are not preceded by a B stimulus (see STAR Methods). This directly avoids the stimulus-specific adaptation that would be present under a BBBBA condition.

It is possible that longer-term adaptation effects in V1 emerging across days, could have altered responses in the Westerberg study, as previously shown.[43] In Westerberg et al., mice viewed the sequences for 3–5 days, over 2,000 times, prior

to testing. In our study, mice saw the sequence only 35 times during a single habituation sequence just prior to testing, and we found that deviance detection to the globally predictable "B" was eliminated within minutes of the initial habituation sequence. This is consistent with LFP studies, where, within a single session, signals in mouse V1 demonstrated complex sequence learning (e.g., ABCD) and deviance detection (e.g., ABDC).[10]

Notably, our single neuron calcium imaging recordings suggest that deviance detection responses (to the global and local oddball C) are present in V1 at the earliest latencies, consistent with our past reports in simple oddball paradigms.[12] A recent theoretical paper[7] argues that such early-latency deviance-detection responses (even to complex sequences, as reported in Knudstrup et al.[10]) are evidence that they reflect local adaptation that is simply fed forward from lower levels, rather than integrative prediction errors. However, another possibility explaining this early latency effect is that top-down modulation in predictable contexts is ongoing/tonic and present in anticipation of visual stimuli, i.e., prior to or timed to the onset of anticipated stimuli.[1] In such a manner, it modulates V1 not in a reactive manner on a trial-by-trial basis, but in an ongoing anticipatory manner, altering excitability with respect to contextually typical features (see Aizenbud et al.[34]).

Consistent with this point, feedback inputs to V1 from mouse ACa do not themselves exhibit deviance detection during standard oddball paradigms, as evidenced by calcium imaging of ACa axonal segments in V1,[1,2] despite the fact that such feedback inputs are nevertheless necessary for deviance detection within V1. This suggests that ACa supports deviance detection not by relaying the deviance detection signal back to V1 but by modulating V1 in an ongoing, tonic fashion to alter context-dependent excitability. In our experiments, ACa's role in modulating V1 was not simply inhibitory or excitatory but was modulatory—in some cases, ACa suppression enhanced V1 activity (Figure 4I) while in others, it suppressed activity (Figures 1G and 1H).

Future work should determine how these modulatory signals interact with local inhibitory microcircuits in V1 and whether disruptions of this modulatory circuitry contribute to altered predictive processing in neuropsychiatric disease.[44] The results presented in Figure 3 carry implications for interpreting a volume of clinical studies on sensory integration in psychiatry.[45] The basic oddball paradigm is ubiquitous in clinical neuroscience research, as several large-scale patient-population studies employ simple oddball sequences for extracting EEG biomarkers of deviance detection.[46,47] Our findings emphasize that the deviance detection signals evoked during these simple paradigms include aspects of higher-order cognition—a point consistent with the fact that simple oddball paradigms correlate with multiple higher-level symptoms and cognitive deficits in schizophrenia.[45,48] Further, dynamic modulations of gain within cortex entail fast synchronous interactions between pyramidal neurons and inhibitory neurons,[1,40,49] which implicates excitation-inhibition balance more broadly, a current focus of schizophrenia research.[50–52] How specific interneuron populations factor into the global vs. local context effects observed here is an important topic for future research.

Limitations of the study

Certain limitations of this study should be noted. First, our design relied on a relatively small number of trials per condition, which allowed us to capture the rapid dynamics of contextual learning (e.g., see Figure S1), but likely limited the precision with which we could estimate the true proportion of deviance-detecting neurons. To address this, we adopted a fixed, uncorrected threshold ($p < .025$) for classifying individual neurons as deviance-detecting via within-neuron deviant-vs-control t tests across trials. This criterion was intended to provide an upper-bound estimate of the number of neurons capable of expressing deviance detection, rather than a precise or conservative count, and the true proportion is likely lower. This choice of using fewer trials also limited our ability to understand how motor variables like locomotion or facial movements could modulate simple vs. complex deviance detection. More broadly, we note a practical tension inherent to the effort of robustly quantifying deviance detection at the single-cell level. Accuracy requires more trials, but increasing trial counts necessarily habituates the very effect being measured.

Second, the total number of neurons recorded per field of view ($n \approx 38$) or per mouse ($n \approx 86$) was constrained by relatively small fields of view and a focus on soma (see STAR Methods). This provides confidence that included neurons reflected genuine somatic activity in layer 2/3 rather than neuropil contamination or drift from other layers, but it did reduce the overall yield of analyzable neurons per animal. We made up for this by increasing our mouse numbers, including LFP recordings as well, and having an internal confirmation (Figure 4) of our main effects, in two additional cohorts of animals (calcium and LFP). Thus, our effects are robust. However, future work with, e.g., mesoscope, multiplane imaging can provide a deeper picture of how larger populations of neurons, across layers and regions, respond to complex deviants.

One notable detail of our primary experiments (Figures 1 and 2) is that we used GCaMP6s. This indicator optimizes detection of activity but has somewhat slower off kinetics compared with other more recent GCaMP variants. Since our paradigm involves rapid presentation of stimuli (1 Hz), this could have compromised our ability to detect certain types of responsive neurons. In the DREADDs experiments shown in Figure 4, we use both GCaMP6s and GCaMP8m ($n = 5$; 2 females), which is a faster variant that can partially mitigate this limitation.[53] As a supplementary analysis, we present our baseline (saline-treated) GCaMP8m data in Figure S5. These results are also presented in Figure S3. Just like in GCaMP6s experiments (Figure 1), we found no deviance detection to the predictable B deviant ($F(1,72) = 1.08$, $p = 0.30$) but strong deviance detection to the global C deviant ($F(1,81) = 18.06$, $p = 5.67e^{-5}$). Again, we did not find deviance detection to the global A deviant ($F(1,71) = 1.08$, $p = 0.30$), but, similar to the LFP results, we also did not find a suppression to this A deviant either. Ultimately, this mixed pattern of effects when the deviant is also a local redundant stimulus suggests that stimulus-specific adaptation may obscure deviance detection in V1.

Finally, our task was entirely passive, as mice were not required to make any response to, or act upon, the stimuli. This design choice allowed us to avoid the confounds of reward

anticipation, to capture rapidly emerging and rapidly suppressed local deviance signals that might otherwise be altered by task engagement, and to characterize sensory processing under unstructured, unmotivated viewing conditions, which likely constitute a substantial portion of natural sensory experience for these animals. However, this also means it remains unclear how attention, motivation, or active engagement with the stimuli (such as would be introduced by a behavioral task requiring responses for reward) might modulate the global and local context effects we describe here. Past work has employed active-discrimination tasks during simple oddball paradigms,[27,28] and so a similar extension to our global deviance-detection condition could provide additional insight.

RESOURCE AVAILABILITY

Lead contact

Requests for further information, resources, and reagents should be directed to and will be fulfilled by the lead contact, Jordan P. Hamm (jordan.hamm@nyulangone.org).

Materials availability

No new reagents or materials were generated for this study.

Data and code availability

- Calcium imaging and electrophysiology data have been deposited at Zenodo and are available at Zenodo: 10.5281/zenodo.21517169 (2P photon calcium imaging) or at Zenodo: 10.5281/zenodo.21517242 (LFP data) and are publicly available as of the date of publication. Accession numbers are listed in the key resources table.
- All necessary code to analyze these data have been deposited at Zenodo: 10.5281/zenodo.22774180.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.P.H., D.S.P., and F.I.; investigation, F.I., G.B., J.M.R., L.R., and M.A.H.; formal analysis, J.P.H.; writing – original draft, J.P.H.; writing – review and editing, all authors; data curation, F.I. and J.P.H.; funding acquisition, J.P.H. and D.S.P.; supervision, J.P.H.

DECLARATION OF INTERESTS

The authors declare no competing interest.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the authors used Claude in order to shorten word count in the abstract section and to refine MATLAB code for statistical analyses. After using this tool or service, the authors heavily reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
AAV1-hSyn-flex-hM4Di-mCherry	Addgene	RRID: Addgene_44362
AAVrg-hSyn-Cre	Addgene	RRID: Addgene_105553
AAV8-CaMKIIa-hM4Di-mCherry	Addgene	RRID: Addgene_50477
AAV9-CaMKIIa-GCaMP8m	Addgene	RRID: Addgene_176751
Chemicals, peptides, and recombinant proteins		
Deschloroclozapine (DCZ)	Sigma-Aldrich	Cat# SML3651
Paraformaldehyde (PFA)	Sigma-Aldrich	Cat# 441244
DAPI	ThermoFisher	Cat# D1306
Antifade mounting medium	Vector Laboratories	Cat# H-1700–2
Tissue-Tek O.C.T. Compound	Electron Microscopy Science	Cat# 62550–01
DiI dye	Sigma-Aldrich	Cat# V22885
Experimental models: Organisms/strains		
Mouse: C57BL/6	Jackson Laboratories	RRID:IMSR_JAX:000664
Mouse: tetO-GCaMP6s;CAG-tTA2 (tm162)	Jackson Laboratories	RRID:IMSR_JAX:031562
Mouse: VGluT-Cre (tm1.1)	Jackson Laboratories	RRID:IMSR_JAX:037512
Software and algorithms		
MATLAB	MathWorks	RRID:SCR_001622
Psychophysics Toolbox	N/A	RRID:SCR_002881
Prairie View	Prairie Technologies/Bruker	N/A
Moco plugin (for ImageJ)	Ref. 53	N/A
ImageJ/Fiji	NIH	RRID:SCR_003070
OASIS (constrained-foopsi)	Ref. 55, 56	N/A
CaImAn	Ref. 55, 56	N/A
Suite2P	Ref. 54	N/A
Custom code for analysis	N/A	10.5281/zenodo.22774180
Deposited data		
Calcium imaging data	N/A	10.5281/zenodo.21517169
Electrophysiology (LFP) data	N/A	10.5281/zenodo.21517242
Other		
Two-photon microscope, Investigator	Bruker Corporation	N/A
Laser, Chameleon Ultra II	Coherent Inc.	N/A
Pockels cell, 350–105 (with 302 RM driver)	Conoptics	N/A
Objective, 16X/0.80W	Nikon	N/A
Bipolar electrode, 2 Channel Electrode	P1 Technologies	Cat# MS303/2-A/SPC
Differential amplifier, DP-304A	Warner Instruments	N/A
Noise cancellation unit, D400	Digitimer	N/A
Belt-treadmill (low-friction, w/rotary encoder)	LABmaker	N/A
LCD monitor, P27P5	Dell	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Experiments were carried out under the guidance and supervision of the Georgia State University (GSU) Division of Animal Resources (protocol number A21066) and the Nathan S. Kline Institute (NKI) Division of Animal Resources (protocol number AP2024-753-NKI). All procedures were approved via Institutional Animal Care and Use Committees (IACUCs) at GSU and NKI. Adult C57BL/6 mice (n = 42, from Jackson Laboratories) were used. Mice were weaned at P25 and group housed (max 5 mice per cage) separated by sex. They were maintained at a 12/12 light/dark cycle (light: 6am-6pm) with food and water available *ad libitum*. Animal ages varied between P73(min) and P132 (max) and averaged at P99 days. Experimental mice were chosen randomly based on availability and age; both female and male mice were included in similar proportions (19F, 23M). For calcium imaging experiments in Figures 1 and 2, transgenic lines were made using mice expressing cre-dependent GCaMP6s (tm162(tetO-GCaMP6s, CAG-tTA2)) crossed with tm1.1-(VGluT) lines (n = 10, 6F). For calcium imaging experiments in Figure 4, we utilized C57BL/6J mice (n = 5, 2F) or GCaMP6s reporter mice (n = 5, 0F). For LFP recordings, we used wildtype mice of C57Bl/6 origin (n = 9, 5F for Figure 1; n = 13, 7F (one excluded because of absent V1 but used for confirming DREADD effects on ACa)). When possible, sex main effects and interactions were examined and not present. For the calcium imaging results in Figure 4, only two female mice were included, so the absence of effects should be interpreted with caution.

METHOD DETAILS

Surgical procedures

For experiments involving calcium imaging, head-plate fixation and craniotomy surgeries were carried out together as previously described.[1,2,11,15] Mice were anesthetized under isoflurane (3% induction, 1–2% maintenance). Anesthesia depth was monitored via toe-pinch. Then a hole with a 3 mm diameter was drilled in the mouse skull in stereotaxically defined left V1 (coordinates from bregma: X = 2 mm, Y = −2.92 mm), followed by the removal of the skull and exposure of brain surface; dura matter was conserved. If viruses were injected (see below), they were administered at this step. Then, a cover glass was placed and sealed at the hole location. Then a titanium head-plate was attached to the mouse head to allow for head-fixation during imaging. After surgeries, mice received analgesics: either 1.0mg/kg Buprenophine (extended release) or Carprofen (5mg/kg, twice daily for three days) subcutaneously.

GCaMP expression was targeted using one of the following the following three methods: (1) vGlut1-Cre; GCaMP6s reporter mice (Figures 1 and 2; n = 10). (2) AAV9-hSyn-Cre (Addgene #105553; titer: 1.0x1012 vg/mL) was injected into the V1 of GCaMP6s reporter mice (100 nL per site; 5 sites per animal; Figure 4, n = 5). (3) AAV9-CaMKIIa-GCaMP8m (Addgene #1176751; titer: 1.0x1012 vg/mL) was injected into the V1 of C57BL/6J mice (100 nL per site; 5 sites per animal; Figure 4, n = 5). For ACa suppression experiments, we either a) injected AAV1-hSyn-flex-hM4Di-mCherry (Addgene #44362, 2.4x1013) in ACa and AAVrg-hSyn-Cre (Addgene #105553, 2.5x1013) in V1 (n = 4) or b) 300nL of AAV8-CaMKIIa-hM4Di-mCherry (Addgene #50477 2.1x1013) in ACa (n = 18) stereotaxically. The following coordinates were used: for ACa, two sites (200nL each), coordinates from bregma: x = 0.3 mm, y = 0.6 and 1.2 mm, z = 0.9 mm), V1: 300nL, coordinates from bregma: x = 2.2 mm, y = −2.9mm, z = 0.5mm).

For local field potential experiments, head-plate attachment was carried out prior to electrode implantation. A bipolar electrode (with contacts spaced approximately 200-600μm apart, Protech international inc. #MS303/2-A/SPC, Insulated wire diameter:0.23mm) was inserted below the dura in stereotaxically defined V1 (coordinates from lambda, X = 2.3mm, Y = 0.7mm) approximately 0.5mm between the dura and white matter. For DREADD experiments, to confirm the impact of DREADD suppression, twelve mice had a second electrode inserted into ACa (Coordinates from bregma, x = 0.3mm, y = 1.0mm, z = 0.9 mm). In four of these mice, ACa targeting was unsuccessful, totaling 8 for statistical comparisons (Figure 4D). Prior to recordings, mice underwent multiple sessions to acclimate them to head-fixation and visual stimulation (oriented gratings), as previously described.[1] Mice only viewed the oddball paradigms on the day of recording. On the day of recording, mice first viewed the regular oddball paradigm, followed by the global/local paradigm.

Visual stimulation

Visual stimulation was presented on a 27 inch flat LCD screen (Dell, P27P5) at a 45° angle from the animal axis, approximately 15 cm from the eye, using Psychophysics Toolbox on MATLAB (Mathworks). Stimuli were full-field, black and white, sinusoidal moving gratings were presented at 100% contrast, 0.08 cycles per degree, two cycles per second, at 8 possible orientations (30°, 45°, 60°, 90°, 120°, 135°, 150°, and 180°).

For the global/local paradigm, short sequences of five stimuli of 500ms in duration, with an inter-stimulus interval of 500-550ms of gray screen, were presented, followed by 6000ms of gray screen (inter-sequence interval). Initially, mice viewed a ‘‘habituation set’’, which included 70 sequences of AAAAB where A was 0° and B was 90°. After this, mice rested for 2–3 min, and then viewed a ‘‘test set’’, wherein AAAAB sequences were primarily presented, but AAAAA (global deviant, local redundant) and AAAAC (global deviant, local deviant) catch trials were presented one in 6 presentations (C was always 45°). Then, after another brief rest, a control run was presented, wherein random orientations (e.g., CDEAB) were presented in the same 5 stimulus sequences (500ms duration, 500-550ms interstimulus interval (ISI), 6000ms inter-sequence interval). A subset of mice then rested 5 min and this set of three paradigms was repeated, where A and B were switched.

For the standard oddball, the same stimulus duration and ISIs were used, but stimuli were presented in continuous trains of 250 trials (≈4 min). This consisted of a "many-standards control" (equally rare, randomly presented stimuli at all 8 possible orientations) and an oddball sequence. The oddball sequence consists of a repetitive sequence of one stimulus ("redundant", either 0° or 90° degree angles, presented 87.5% of the time), randomly interrupted by a stimulus of a different orientation ("deviant", 120°, 135°, or 180° degree angles, presented 12.5% of the time). In the latter half of the sequence, the redundant stimulus is "flipped" to become the deviant, and vice versa ("oddball flip").

Two-photon calcium imaging

Two-photon microscopy (28 Hz framerate; Bruker Investigator laser scanning two-photon microscope; Bruker Corporation, Billerica, MA, USA) excited by a laser (Chameleon Ultra II, Coherent Inc, Santa Clara, CA, USA) at 920 nm wavelength was used to image the fluorescent calcium sensor GCaMP6s expressed in excitatory neurons in the mouse visual cortex. The laser beam was modulated with a Pockels cell (Conoptics 350–105, with 302 RM driver) and scanned with galvometers through a water immersion objective (16X/0.80W, Nikon, Tokyo, JP). The objective lens was positioned over the animal's head while a small volume of Aquasonic ultrasound gel (Parker Laboratories Inc) was placed at the site of the cranial window to bridge the objective with the imaging area in the sessions. The animals were awake, head-fixed under the microscope by their headplate, while sitting on top of a treadmill (low-friction mouse belt-treadmill with rotary encoder, from LABmaker, Berlin, Germany) free to move forward and backwards. All recordings were carried out in a dark room with the researcher present to monitor mouse wakefulness and check for signs of discomfort. Each run had a duration between 4 and 13 min. Scanning and imaging were done through Prairie View (Prairie Technologies) software (8 KHz resonant galvanometer, recorded at ∼56 frames per second, downsampled to 28 frames per second by averaging two-frames, for 256x256×pixels, 2× magnification, 401× 401 μm field of view, 1.57 μm pixel size). A time-series was recorded using Prairie View software as the mice observed visual stimuli. The visual stimulus was transmitted to the monitor through an HDMI cable, converted to a voltage trace, and connected to the computer through the Voltage Recording tool on Prairie View. Time-series and stimulus voltage traces were synchronized at the onset of recording for proper alignment of neuronal activity and stimulus presentation. Images were taken 100 μm–350 μm from the brain surface, approximately L2/3 of the mouse cortex. Locomotion was recorded during two-photon calcium imaging via a rotary encoder integrated into the treadmill.

Local field potentials recordings

For the LFP experiments, the mouse was fixed to the recording apparatus through the head-plate and free to locomote on the treadmill. Insulated cables were connected to the implanted electrodes and plugged into a differential amplifier (Warner instruments, DP-304A, high-pass: 0 Hz, low-pass: 500 Hz, gain: 1K, Holliston, MA, USA). Amplified signals were sampled at 1kHz and were passed through a 60 Hz noise cancellation machine (Digitimer, D400, Mains Noise Eliminator, Letchworth Garden City, UK), which, instead of filtering, creates an adaptive subtraction of repeating signals which avoids phase delays or other forms of waveform distortion. Prior to insertion, probes were submerged in DiI dye for post-hoc anatomical validation. Electrophysiological activity was recorded using the Prairie View software. Visual stimulus timings were recorded as voltage traces at the same time as the LFPs signals for proper alignment. Locomotion was recorded during electrophysiology via a rotary encoder integrated into the treadmill.

Chemogenetic suppression of ACa

On the day of recording, mice were injected with DCZ (Sigma-Aldrich, SML3651, final concentration 0.1mg/kg)[54] or saline I.P and fixed to the treadmill. Ten minutes later, mice viewed global-local habituation, global-local oddball, and control runs as described above. In 2P-Ca^{++} recordings, mice viewed an additional set after reversing the A and B stimuli (becoming BBBBA), during which another field of view was recorded. Then mice were returned to their home cage. Four days later, they were injected with the opposite drug (i.e., saline if they received DCZ on day 1, or vice versa) and recordings were repeated.

Histological verification

Mice were deeply anesthetized via urethane and 3% isoflurane and then perfused transcardially with PBS followed by 4% paraformaldehyde (PFA)/PBS. Heads with electrodes and headplates attached were post-fixed in 4% PFA/PBS for two overnights. After isolation, brains were cryoprotected with 30% sucrose/PBS for one to two overnights and then embedded in Tissue-Tek OCT compound (Sakura) and cryosectioned (50 μm). Sections were stained with DAPI (ThermoFisher, D1306, 1μg/ml) and mounted using antifade medium (VectorLabs, H-1700–2).

QUANTIFICATION AND STATISTICAL ANALYSIS

2-Photon image processing and analysis

Calcium imaging datasets were scored similarly to previous reports.[15] Raw images were processed to correct motion artifacts using the "Moco" plugin for ImageJ (NIH).[55] Regions of interest (ROIs) were selected semi-automatically from time-averaged images using an in-house written MATLAB (MathWorks) routine or with Suite 2P.[56] Individual fluorescence traces were visually inspected and kept if they resembled soma (>5μm in diameter) and showed at least one calcium transient in both runs (control and oddball), and no evidence of dramatic drift. For Figures 1 and 2, this yielded 36.0 neurons per field of view on average. For Figure 4, this yielded

42.1 neurons per field of view. Fluorescence traces were subtracted from the pixels just outside these ROIs (i.e., halo subtraction). Because of the relatively rapid stimulus presentation used in this study ($\approx$1Hz) relative to the decay time of GCaMP calcium indicators, neural activity was analyzed by deconvolving putative spikes from fluorescent traces using a constrained non-negative matrix factorization algorithm (constrained-foopsi,[57] available in the Oasis or Calman toolboxes[58]) and smoothing the result with a 140ms Gaussian window (5 samples) to estimate firing rate (eFR). For analyses, eFR was standardized (eFRstd) using the standard deviation of the bottom 50% of positive values across a given recording. Standardizing did not impact the statistical conclusions or the pattern of the effects, but helped normalize values for clearer visualization across neurons.

For the global-local paradigm, eFRstd was averaged across trials for each stimulus type (control-A,B,C; deviant-A,B,C; redundant-A). As each mouse saw just 6 of the type A and C deviants, we limited analyses to 6 trials of each of these conditions. For the control run, we only analyzed A, B, or C orientation trials i) if they fell in the 4th or 5th spot in the sequence, and ii) if they were not an immediate repeat (i.e., a C stimulus in the 3rd/4th spot). For A and B stimuli during the testing run, we analyzed trials in the sequence immediately preceding the catch sequences. That is, the 4th A or 5th B response in AAAA-B that comes right before an AAAA-C. For the standard oddball paradigm, we also limited to 6 trials in order to facilitate optimal interparadigm comparisons. We focused on the first 6 deviants of each orientation and the final 6 instances of a control orientation during the control run (as the control sequence was always played first, this helped keep the trials being compared closer in time). Further, controls were excluded if they were repeats with the previous trial, as described above. Trials with detectable locomotion detected via rotary encoder (see above) were excluded (<15% of trials for any mouse). Specifically, if locomotion traces were visually inspected and thresholded above the noise floor, which was typically around 40 mVs, and if any treadmill movement occurred within 1 s of stimulus onset or offset, a trial was excluded.

We focused analysis on cells with significant stimulus-evoked responses to a given orientation. This was determined based on the following criteria: either i) average responses >1.97 standard deviations ($p < .05$, two-tailed) above its pre-stimulus baseline to both conditions (control and deviant), or ii) average responses >2.6 ($p < .01$) above its pre-stimulus baseline to either condition. The standard deviation was estimated based on a cell's activity levels across all trials. Importantly, these choices of cell inclusion had no tangible effects on the conclusions (see Figure 2), but were chosen to allow for the most accurate effect size estimates and consistency with past studies.

Note on data inclusion: Each mouse completed three runs (training, test (oddball run), and control) for forward (A = 90 deg, B = 0 deg, C = 45 deg) and then reverse orientation pairings (A = 0 deg, B = 90 deg, C = 45 deg). We included a dataset (forward or reverse) in the main analysis if i) a mouse had an average of 6 trials (for A, B, and C deviants) across the test and control runs and a minimum of 4 trials for any stimulus or run included after excluding trials due to locomotion and other movement artifacts, and ii) there were greater than 10 usable neurons present. As trial numbers were generally low (to maintain true surprise for deviants), this led to us only use both forward and reverse runs for 3 mice (out of 10) in the initial analysis (Figures 1 and 2), 8 mice out of 10 for the saline condition (Figure 4), and 5 out of 10 mice for the DCZ condition (Figure 4). Notably, the sal/DCZ experiments had slightly more datasets included than the initial dataset (Figures 1 and 2) due to the fact that we doubled the number of control trials in this set of follow-up experiments, specifically to account for this issue. The number of training and test trials remained the same. For the training run, we only collected usable calcium data for 5 mice, but given the large number of repeats in these conditions, both directions were usable for all 5.

Local field potential signal processing and analysis

Trials with excessive signal (> $\approx$ 5 std devs) in V1 were manually excluded (between 0 and 20). All analyses were limited to the same trial choices as described above. Ongoing data were converted to the time-frequency domain with a modified morelet wavelet approach with 79 evenly spaced wavelets from 4 to 100Hz, linearly increasing in length from 1 to 20 cycles per wavelet, applied every 5ms from 300ms pre-to 700ms post stimulus onset (200ms post-stimulus offset) as previously described.[2] Stimulus-induced power spectra were computed for all conditions for each mouse and baseline-corrected by subtracting the average for each frequency in the 100ms prior to stimulus onset. Based on past work,[1,12] we compared control vs. deviant responses in theta (4-12Hz) and gamma power (20-40Hz) and the 75 to 600ms time-period. For analysing whether DREADDs impacted ACa, we analyzed baseline power. We segmented LFP recordings during the control and oddball runs, focusing on 1 s periods with no locomotion (see above) or visual stimulation. We equated the number of segments across all mice ($n = 245$). We carried our fast Fourier transforms, taking the natural log of the power (squared amplitude) for each segment, and then the median value for each frequency (4–200 Hz) across segments, for each mouse, condition (Sal, DCZ), and region (V1 vs. ACa). Analysis focused on high gamma power, 80 to 200-Hz. Histological verifications indicated that all mice except one had V1 successfully targeted (for a total of 12 mice for Figure 4). Additionally, 8 mice had ACa successfully targeted.

Statistics

Complete statistical details of all experiments, including statistical tests used, exact value of n, what n represents (e.g., number of animals or number of cells) can be found in Table S1. Key information is also presented throughout the text in figure legends and the Results section. Meaning of asterisks (*) are found in the figure legends, along with the associated statistical test, as well as in Table S1. All bar and line plots represent means and error bars represent standard error of the mean, either across neurons (for calcium imaging) or across mice (for local field potentials and behavior). All statistics were carried out in MATLAB, using publicly

available code (10.5281/zenodo.22774180). For calcium imaging analyses, we focused on cell-wise analyses and carried out a linear mixed-effects models (LME) to account for the fact that many neurons came from the same mice (mouse as random-effects), and some fields of view (FOVs) came from the same mice. In these models, significant effects of deviant vs. control were the primary focus, and were reported as F-values, with satterthwaite-estimated degrees of freedom, which are used to estimate the true df in nested LMEs. For LFP experiments in Figure 1, only one set of values came from each mouse, so we carried out factorial repeated-measures ANOVAs on induced power at the mouse level with CONTEXT (control, oddball) and DEVIANT-TYPE (redundant, (A) local-redundant, global-deviant, (B) local-deviant, global-redundant, (C) local deviant, global deviant) as within-subjects factors. Significant interactions were tested further with paired-samples t-tests. For DREADDs experiments (Figure 4), we focused statistical tests on deviant types B vs. C, as this is where our main effects were in Figures 1 and 2. As some mice received saline first while some received DCZ first, we included an ORDER variable as a fixed effects factor (Day 1 vs. Day 5) in LFP and 2P-Ca^{++} experiments. For LFP, carried out linear mixed effects models on induced power at the mouse level with CONTEXT (control, oddball) and DEVIANT-TYPE ((B) local-deviant, global-redundant, (C) local deviant, global deviant) and DRUG (SALINE, DCZ) as fixed effects variables and mouse as random effects variable. We tested specifically for a 3 way interaction of context, deviant type, and drug conditions. For the 2P-Ca^{++} DREADD experiments, we used a similar model, but focused on individual neurons and included mouse, field of view (FOV), and cell ID as additional, nested random effects variables. Additionally, indicator (GCaMP6s vs. 8m) was included as a fixed-effect. For analysing baseline power in LFP recordings, we used a linear mixed effects model analyzing drug condition (SALINE, DCZ) and including order as a fixed effect. Sex was included as a fixed effect in all models. As an exploratory step, we explored interactions with sex, but observed no significant relationships. To follow-up on significant interactions, we carried out t-tests (one or two-tailed, depending on the condition) with Benjamini-Hochberg False Discovery Rate correction at alpha = .05 to control for multiple comparisons.[59]